

Higher Gauge Theories Based on 3-groups

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Abstract

We study the *categorical generalizations* of a BF theory to $2BF$ and $3BF$ theories in the framework of higher gauge theory. We construct the constrained $3BF$ actions describing the correct dynamics of Yang-Mills, Klein-Gordon, Dirac, Weyl, and Majorana fields coupled to Einstein-Cartan gravity. The action is naturally split into a topological sector and a sector with simplicity constraints, adapted to the *spinfoam quantization programme*. **The structure of the 3-group gives rise to a novel gauge group which specifies the spectrum of matter fields present in the theory**, just like the ordinary gauge group specifies the spectrum of gauge bosons in the Yang-Mills theory. This allows us to rewrite *the whole Standard Model coupled to gravity as a constrained 3BF action*, facilitating the nonperturbative quantization of both gravity and matter fields.

Category theory

a Lie crossed module $(H \xrightarrow{\partial} G, \triangleright)$

- Lie groups G and H ,
- a homomorphism δ from H to G ,
- an action of G on $H \triangleright : G \times H \rightarrow H$.

a Lie 2-crossed module

$$(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _ \})$$

- Lie groups G , H and L ,
- homomorphisms δ and ∂ ,
- an action $\triangleright$ of the group G on all three groups,
- the Peiffer lifting $\{_, _ \} : H \times H \rightarrow L$.

Conclusions

- The scalar and fermion fields can be specified using the third gauge group L .
- Gauge groups corresponding to the Klein-Gordon, Dirac, Weyl and Majorana fields are specified and the relevant constrained $3BF$ actions that describe all these fields coupled to Einstein-Cartan gravity in the standard way are constructed.
- ***The first step of the spinfoam quantization programme for the complete theory of gravity and matter is completed.***
- The presence and the properties of the new gauge group open up a possibility of a non-trivial unification of all fields and a possible explanation of fermion families and all other structure in the matter spectrum of the theory.

References

- [1] T.Radenković and M. Vojinović, [arXiv:1904.07566](#).

Results

- **$2BF$ topological action and $3BF$ topological action:**

$$S_{2BF} = \int_{\mathcal{M}_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}}, \quad S_{3BF} = \int_{\mathcal{M}_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}} + \langle D \wedge \mathcal{H} \rangle_{\mathfrak{l}}.$$

- **Gravity and $SU(N)$ Yang-Mills theory**

– A crossed-module $(H \xrightarrow{\partial} G, \triangleright)$:

1. $G = SO(3, 1) \times SU(N)$, $H = \mathbb{R}^4$,
2. $M_{ab} \triangleright P_c = [M_{ab}, P_c]$, $\tau_I \triangleright P_a = 0$,
3. $\partial(\tau_I) = 0$.

– The 2-connection (α, β) : $\alpha = \omega^{ab} M_{ab} + A^I \tau_I$, $\beta = \beta^a P_a$.

– The 2-curvature $(\mathcal{F}, \mathcal{G})$: $\mathcal{F} = R^{ab} M_{ab} + F^I \tau_I$, $\mathcal{G} = \nabla \beta P_a$.

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$$S_{2BF} = \int_{\mathcal{M}_4} B^{ab} \wedge R_{ab} + B^I \wedge F_I + e_a \wedge \nabla \beta^a.$$

– The constrained action:

$$S = \int_{\mathcal{M}_4} B^{ab} \wedge R_{ab} + B^I \wedge F_I + e_a \wedge \nabla \beta^a - \lambda_{ab} \wedge \left(B^{ab} - \frac{1}{16\pi l_p^2} \varepsilon^{abcd} e_c \wedge e_d \right) + \lambda^I \wedge \left(B_I - \frac{12}{g} M_{abI} e^a \wedge e^b \right) + \zeta^{abI} \left(M_{abI} \varepsilon_{cdef} e^c \wedge e^d \wedge e^e \wedge e^f - g_{IJ} F^J \wedge e_a \wedge e_b \right).$$

- **Real Klein-Gordon field** $D = \phi \mathbb{I}$

– A 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _ \})$:

1. $G = SO(3, 1)$, $H = \mathbb{R}^4$, $L = \mathbb{R}$,
2. $M_{ab} \triangleright P_c = [M_{ab}, P_c]$, $M_{ab} \triangleright T_A = 0$,
3. $\partial(P_a) = 0$, $\delta(T_A) = 0$, $\{P_a, P_b\} = 0$.

– The 3-connection (α, β, γ) : $\alpha = \omega^{ab} M_{ab}$, $\beta = \beta^a P_a$, $\gamma = \gamma \mathbb{I}$.

– The 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$: $\mathcal{F} = R^{ab} M_{ab}$, $\mathcal{G} = \nabla \beta^a P_a$, $\mathcal{H} = d\gamma$.

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$$S_{3BF} = \int_{\mathcal{M}_4} B^{ab} \wedge R_{ab} + e_a \wedge \nabla \beta^a + \phi d\gamma.$$

– The constrained action:

$$S = \int_{\mathcal{M}_4} B^{ab} \wedge R_{ab} + e_a \wedge \nabla \beta^a + \phi d\gamma - \lambda_{ab} \wedge \left(B^{ab} - \frac{1}{16\pi l_p^2} \varepsilon^{abcd} e_c \wedge e_d \right) + \lambda \wedge \left(\gamma - \frac{1}{2} H_{abc} e^a \wedge e^b \wedge e^c \right) + \Lambda^{ab} \wedge \left(H_{abc} \varepsilon^{cdef} e_d \wedge e_e \wedge e_f - d\phi \wedge e_a \wedge e_b \right) - \frac{1}{2 \cdot 4!} m^2 \phi^2 \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d.$$

- **Weyl spinor fields** $D = \psi_{\alpha} P^{\alpha} + \bar{\psi}^{\dot{\alpha}} P_{\dot{\alpha}}$

– A 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _ \})$:

1. $G = SO(3, 1)$, $H = \mathbb{R}^4$, $L = \mathbb{R}^4(\mathbb{G})$,
2. $M_{ab} \triangleright P_c = [M_{ab}, P_c]$, $M_{ab} \triangleright P^{\alpha} = \frac{1}{2} (\sigma_{ab})^{\alpha}_{\beta} P^{\beta}$, $M_{ab} \triangleright P_{\dot{\alpha}} = \frac{1}{2} (\bar{\sigma}_{ab})^{\dot{\beta}}_{\dot{\alpha}} P_{\dot{\beta}}$,
3. $\partial(P_a) = 0$, $\delta(T_A) = 0$, $\{P_a, P_b\} = 0$.

– The 3-connection (α, β, γ) : $\alpha = \omega^{ab} M_{ab}$, $\beta = \beta^a P_a$, $\gamma = \gamma_{\alpha} P^{\alpha} + \bar{\gamma}^{\dot{\alpha}} P_{\dot{\alpha}}$.

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$$\mathcal{F} = R^{ab} M_{ab}, \quad \mathcal{G} = \nabla \beta^a P_a, \quad \mathcal{H} = (d\gamma_{\alpha} + \frac{1}{2} \omega^{ab} (\sigma^{ab})^{\beta}_{\alpha} \gamma_{\beta}) P^{\alpha} + (d\bar{\gamma}^{\dot{\alpha}} + \frac{1}{2} \omega_{ab} (\bar{\sigma}^{ab})^{\dot{\alpha}}_{\dot{\beta}} \bar{\gamma}^{\dot{\beta}}) P_{\dot{\alpha}} \equiv (\vec{\nabla} \gamma)_{\alpha} P^{\alpha} + (\vec{\nabla} \bar{\gamma})^{\dot{\alpha}} P_{\dot{\alpha}}.$$

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$$S_{3BF} = \int_{\mathcal{M}_4} B^{ab} \wedge R_{ab} + e_a \wedge \nabla \beta^a + \psi^{\alpha} \wedge (\vec{\nabla} \gamma)_{\alpha} + \bar{\psi}_{\dot{\alpha}} \wedge (\vec{\nabla} \bar{\gamma})^{\dot{\alpha}}.$$